# Xiao Chen

Ph.D. Student in Computer Science, Indiana University Bloomington

**xc56@iu.edu** | **chenxiao2402.github.io** | **GitHub: chenxiao2402**

**Research interests:** Programming languages; gradual and migratory typing; operational semantics; algebraic effect handlers; formal verification.

Seeking Ph.D. opportunities outside the United States for 2027 entry.

## Education

**Indiana University Bloomington** Sep. 2024 – Present
Ph.D. in Computer Science | Advisor: Prof. Sam Tobin-Hochstadt
Currently on leave due to prolonged U.S. visa processing delays.

**Northwestern University** Sep. 2022 – Jun. 2024
M.S. in Computer Science | GPA: 3.957/4.00
Research with Profs. Christos Dimoulas and Simone Campanoni.

**Nanjing University** Sep. 2020 – Jun. 2022
M.Eng. in Electronic Information | GPA: 4.58/5.00

**Nanjing University** Sep. 2016 – Jun. 2020
B.Eng. in Software Engineering | GPA: 4.28/5.00

## Ongoing Research Projects

**Type as Obligations (TAO)**
*Formalized proofs in Rocq complete; manuscript in preparation.*

- Developed an executable operational framework that separates the type obligations accumulated during execution from the checks performed by a runtime enforcement strategy.
- Mechanically derived a profiler semantics that tracks typing histories and boundary provenance through higher-order interactions, without manually instrumenting individual programs.
- Proved a bidirectional correspondence between ordinary and profiler evaluation: well-typed, uniquely labeled programs terminate with the same resolved values, despite extra bookkeeping and different location allocations.
- Defined obligation soundness and proved that it implies type soundness, but not conversely. Used violated obligations as a common reference for blame soundness and completeness.
- Compared Natural, Threesomes, Forgetful, and Transient using uniform criteria, with formalized proofs and counterexamples in **Rocq/Coq** and an executable **Racket/Redex** model.

**Migratory Typing for Algebraic Effects and Handlers**
*Effect-Typed-Racket / Prototype implemented; formal semantics and proofs in progress.*

- Extended **Typed Racket** with algebraic effect handlers and effect typing, carrying effect information through higher-order calls, polymorphic instantiation, and module imports.
- Implemented runtime contracts for escaping-effect specifications across typed/untyped boundaries. Distinguished forwarded requests and resumed computations from effects of the handler implementation.
- Supported effect identity across modules and explicit effect-row instantiation. Current contracts check concrete effect allowances; relational effect-row parametricity remains ongoing work.
- Developing a high-level calculus and a lower-level model based on delimited continuations and contracts, aiming to establish type and effect soundness and a behavioral correspondence between them.

## Honors and Selected Coursework

**National Scholarship for Postgraduates**, Nanjing University (top 1%, Oct. 2021).
**First-class People's Scholarship** (top 5%, Nov. 2019); **Third-class People's Scholarship** (top 20%, Nov. 2017 and Nov. 2018), Nanjing University.

## Work Experience

**Tencent Holdings Ltd.** Jul. 2021 – Jan. 2022
iOS Development Intern, Technology Engineering Group | Shenzhen, China

- Developed an iOS multimedia messaging app using 5G connectivity; designed messaging card styles and interaction models, and refactored the user guide and network error detection logic.

**Microsoft Software Technology Center Asia** Jul. 2019 – Oct. 2019
Software Engineering Intern, Speech Recognition Group | Suzhou, China

- Automated speech recognition model training for conference scenarios using Microsoft AEther. Applied Gaussian blur and SG filtering to improve recognition accuracy in short conversations.

## Skills and Languages

**Programming and formal methods:** Racket, Rocq/Coq, Redex, Agda, C++, Haskell, Python, Swift, LLVM IR.
**Tools:** Linux shell, Vim, LaTeX, Git, HTCondor.
**Languages:** Chinese (native), English (proficient), Japanese (JLPT N1), French (beginner).